

Chapitre 3 : Les statistiques

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1 Avant de commencer

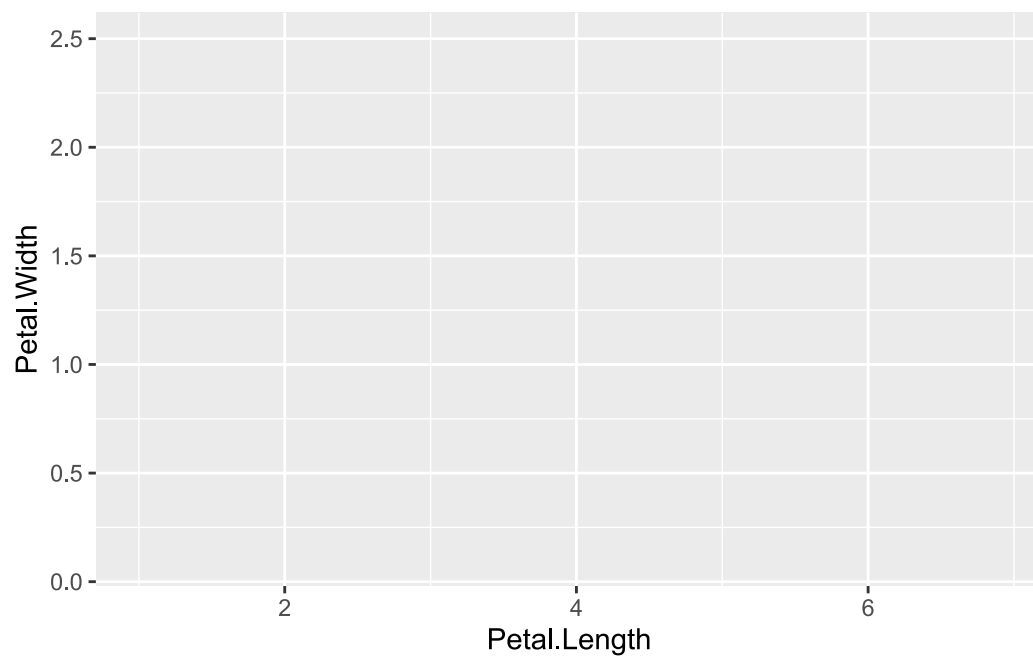
Ce chapitre reprends les codes du chapitre 3 sur *les statistiques* du livre **Langage R et Statistiques** : *Initiation à l'analyse de données* publié par les éditions ENI (2^{ème} édition).

```
library(tidyverse)
```

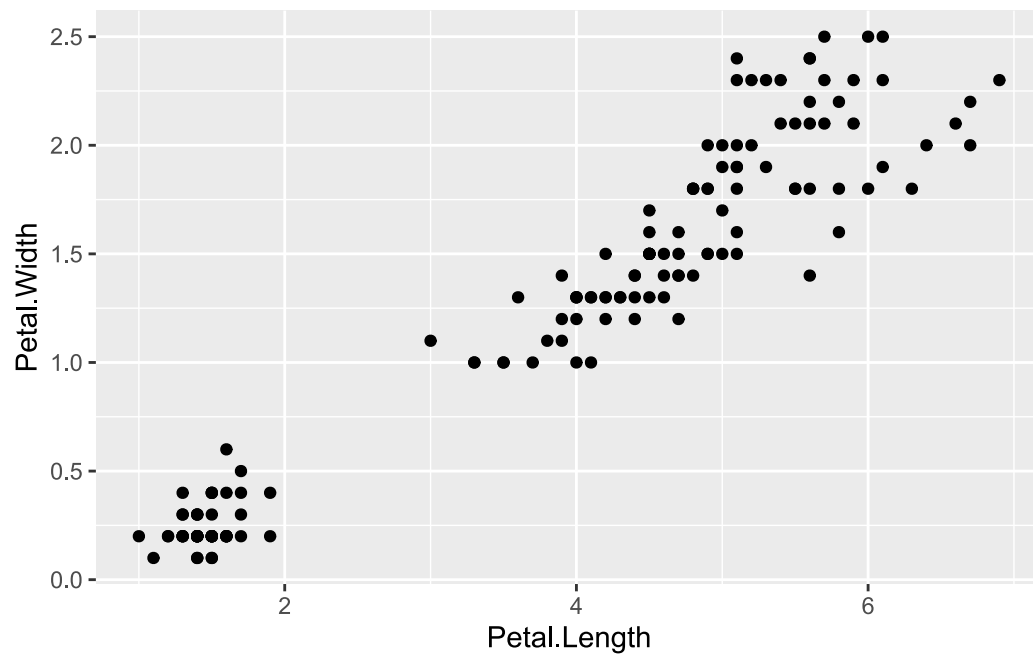
```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1      ✓ readr      2.2.0
✓ forcats    1.0.1      ✓ stringr    1.6.0
✓ ggplot2    4.0.3      ✓ tibble     3.3.1
✓ lubridate  1.9.5      ✓ tidyr      1.3.2
✓ purrr      1.2.2
— Conflicts — tidyverse_conflicts() —
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
  conflicts to become errors
```

2 Prise en main {ggplot2}

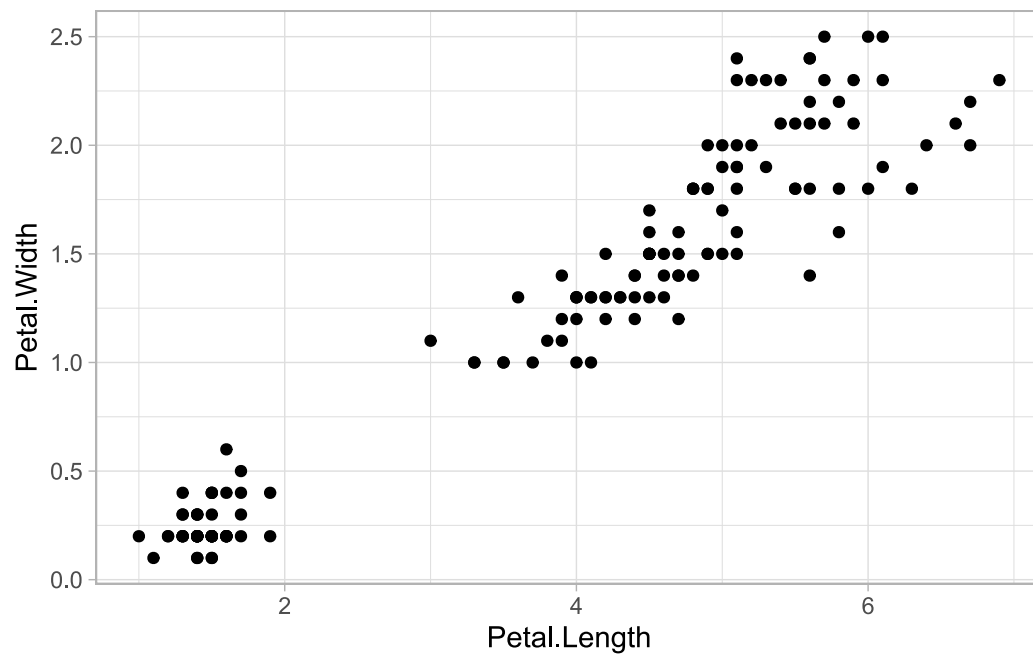
```
ggplot(
  data = iris,
  mapping =
    aes(
      x = Petal.Length,
      y = Petal.Width
    )
)
```



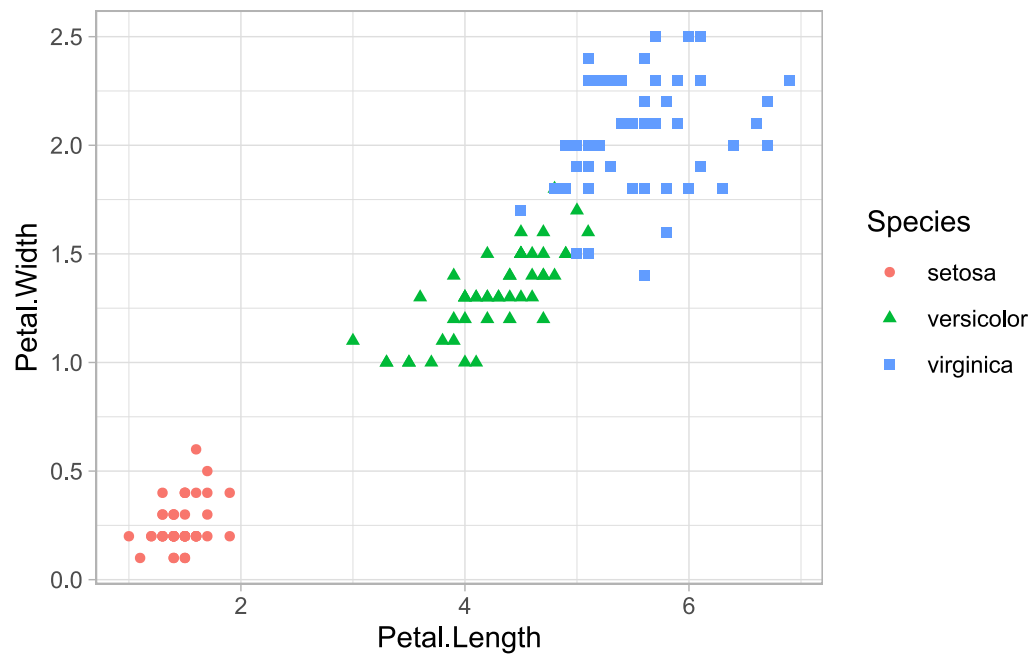
```
ggplot(  
  data = iris  
) +  
  aes(  
    x = Petal.Length,  
    y = Petal.Width  
  ) +  
  geom_point()
```



```
ggplot(  
  data = iris  
) +  
  aes(  
    x = Petal.Length,  
    y = Petal.Width  
  ) +  
  geom_point() +  
  theme_light()
```

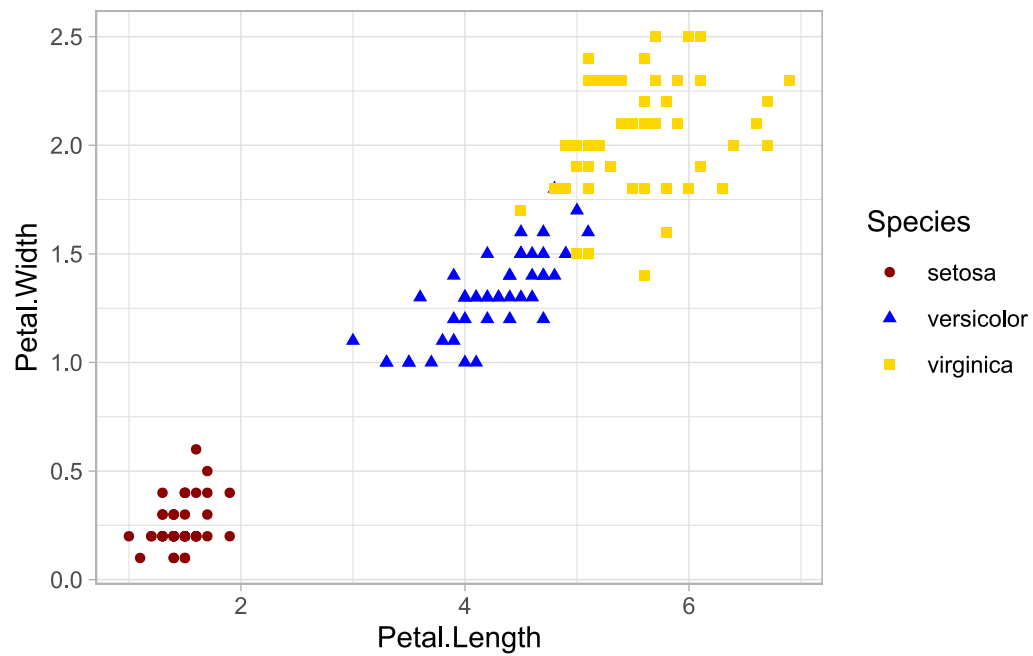


```
ggplot(  
  data = iris  
) +  
  aes(  
    x = Petal.Length,  
    y = Petal.Width,  
    color = Species,  
    shape = Species  
  ) +  
  geom_point() +  
  theme_light()
```



```
ggsave("img/03Rl04N.png", width = 14, height = 10, units = "cm")
```

```
ggplot(  
  data = iris  
) +  
  aes(  
    x = Petal.Length,  
    y = Petal.Width,  
    color = Species,  
    shape = Species  
  ) +  
  geom_point() +  
  theme_light() +  
  scale_color_manual(  
    values = c("darkred", "blue", "gold")  
  )  
)
```



```
vecteur_couleur <- c(
  "setosa" = "#0E7175FF",
  "versicolor" = "#FD7901FF",
  "virginica" = "#C35BCAFF"
)

vecteur_forme <- c(
  "setosa" = 15,
  "versicolor" = 16,
  "virginica" = 17
)

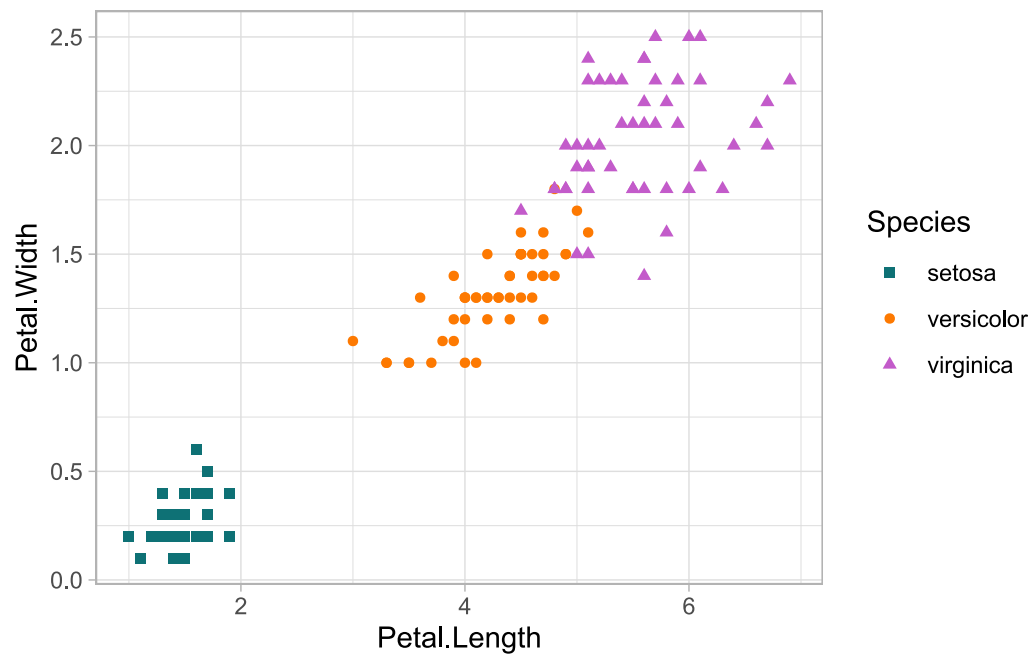
ggplot(
  data = iris
) +
  aes(
    x = Petal.Length,
    y = Petal.Width,
    color = Species,
    shape = Species
  ) +
  geom_point() +
  theme_light() +
  scale_color_manual(
```



```

    values = vecteur_couleur
  ) +
  scale_shape_manual(
    values = vecteur_forme
  )

```



```

ggsave("img/03Rl05Nbis.png", width = 14, height = 10, units = "cm")

```

3 Lois de probabilités

3.1 Initiation loi de probabilités : lancer d'une pièce

Pour illustrer le comportement des lois de probabilités autour des grands nombres, je vais représenter le nombre de faces pour un nombre croissant de lancers de pièces.

3.1.a création d'un vecteur contenant le nombre de lancers

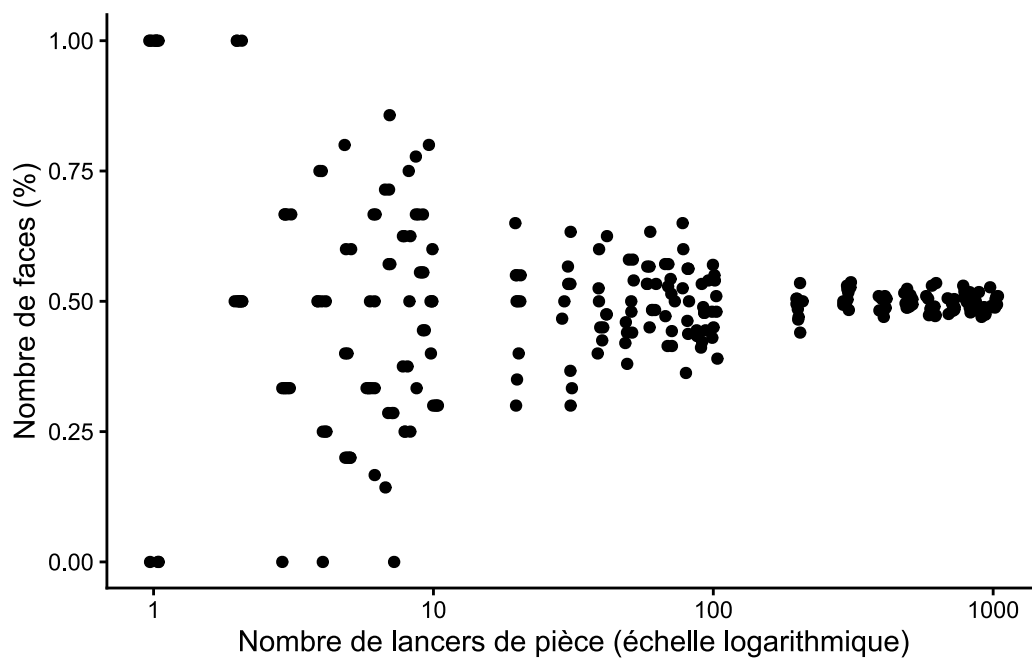
```

nb_lancers <- c(
  1:9, # 1, 2, 3... 9
  seq(from = 10, to = 100, by = 10), # 10, 20... 100
  seq(from = 200, to = 1000, by = 100) # 200, 300... 1000
)

```

3.1.b 10 simulations pour chaque valeurs de nb_lancers de pièces

```
purrr::map(
  c(1:10),
  ~ bind_rows(
    bind_cols(
      nb_lancers = nb_lancers,
      nb_lancers |>
        purrr::map(
          ~ rbinom(.x, 1, 0.5)
        ) |>
        purrr::map(sum) |>
        purrr::map(as_tibble) |>
        bind_rows() |>
        rename(nb_faces = value)
    )
  ) |>
  bind_rows() |> # mise en commun des lignes
  mutate(
    pourcentage_face = nb_faces / nb_lancers # calcul du pourcentage de réussites
  ) |> # création du graphique
  ggplot() +
  aes(x = nb_lancers, y = pourcentage_face, group = nb_lancers) +
  geom_jitter() + # ajout des points
  scale_x_log10() + # choix d'une échelle logarithmique
  theme_classic() +
  labs(
    x = "Nombre de lancers de pièce (échelle logarithmique)",
    y = "Nombre de faces (%)"
  )
)
```



```
ggsave("img/03Rl06N.png", width = 14, height = 10, units = "cm")
```

3.2 Loto biaisé

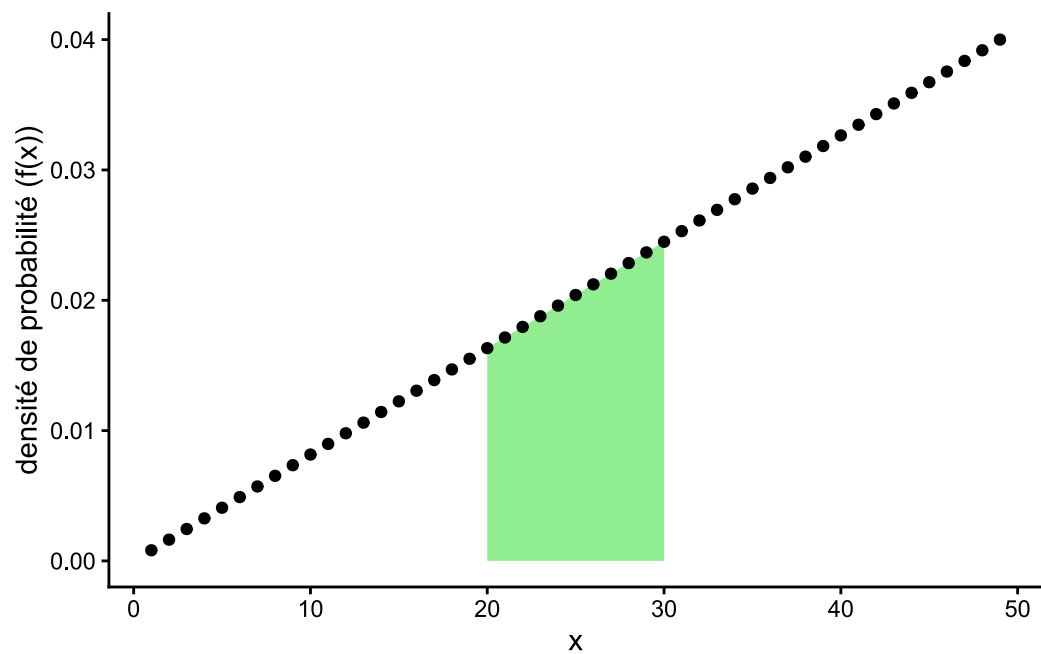
Pour illustrer, voici l'exemple d'un jeu de loto biaisé où les boules de numérotées de 1 à 49 ont un poids proportionnel à leurs chiffres donc la boule 49 aurait une masse 49 fois supérieure à la boule 1

```
# création de la table contenant les données de loto biaisé puis du graphique
tibble(
  evenement = c(1:49), # numéro de la boule 1/49
  densite = c(1:49)/sum(1:49) # probabilité de sortie
) |>
ggplot() +
  aes(
    x = evenement,
    y = densite
  ) +
  geom_polygon( # ajout du polygone vert/gris représentant la probabilité
d'obtenir une boule ayant un numéro supérieur ou égale à 20 mais inférieure ou
égale à 30
    data = tibble(
      x = c(20, 20, 30, 30),
      y = c(0, 20/sum(1:49), 30/sum(1:49), 0)
    )
  )
```

```

    ),
    aes(x = x, y = y),
    fill = "light green"
  ) +
  geom_point() + # ajout de la probabilité de chaque boule de sortir
  theme_classic() +
  xlab("x") +
  ylab("densité de probabilité (f(x))")

```



```

ggsave("img/03Rl07N.png", width = 14, height = 10, units = "cm")

```

3.3 Loi uniforme

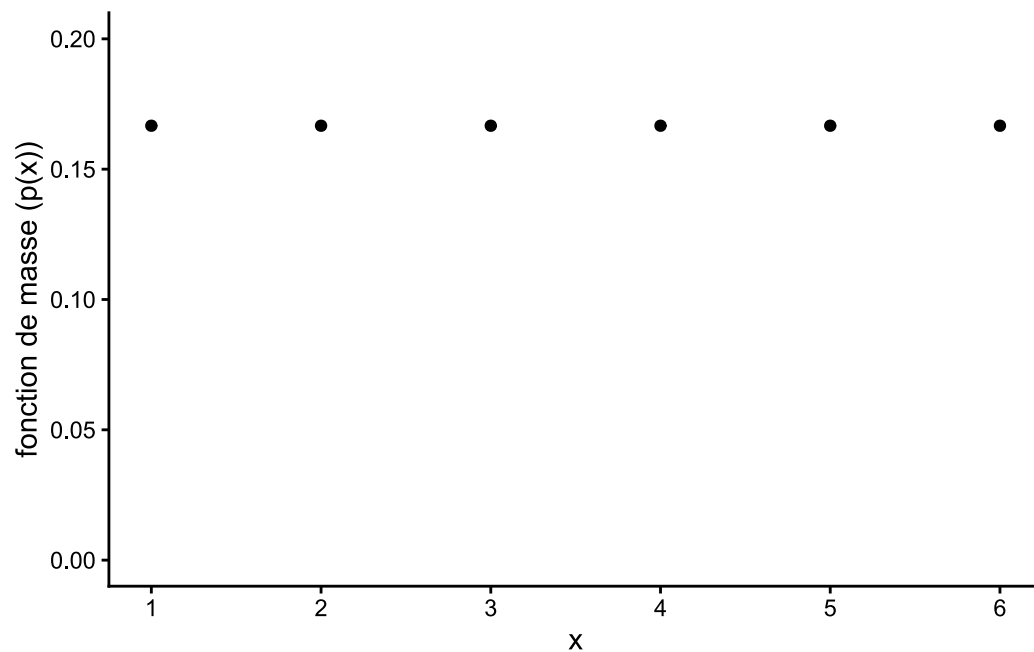
3.3.a pour un dé équilibré

```

tibble(
  evenement = c(1:6),
  probabilite = 1/6
) |>
ggplot() +
  aes(
    x = evenement,
    y = probabilite
  ) +

```

```
geom_point() +
theme_classic() +
scale_x_continuous(breaks = c(1:6)) +
coord_cartesian(ylim = c(0, 0.2)) +
xlab("x") +
ylab("fonction de masse (p(x))")
```



```
ggsave("img/03Rl08N.png", width = 14, height = 10, units = "cm")
```

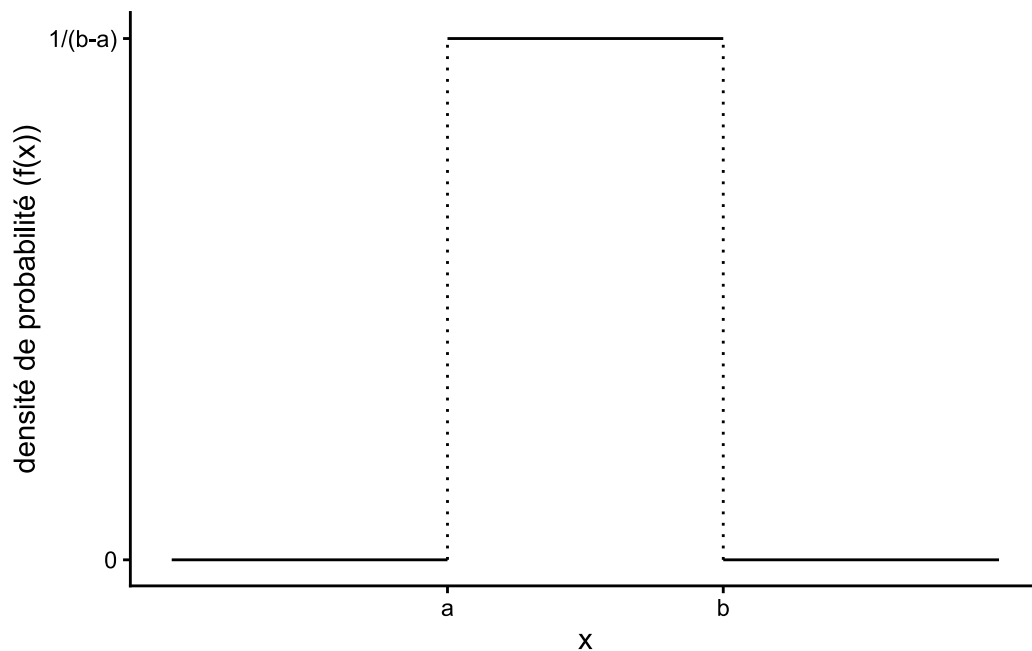
3.3.b loi de densité uniforme continue

```
tibble(
  evenement = c(0:1),
  densite = 1
) |>
ggplot() +
aes(
  x = evenement,
  y = densite
) +
annotate("segment", x = -1, y = 0, xend = 0, yend = 0) +
annotate("segment", x = 0, y = 1, xend = 1, yend = 1) +
annotate("segment", x = 1, y = 0, xend = 2, yend = 0) +
```

```

annotate("segment", x = 0, y = 0, xend = 0, yend = 1, linetype = "dotted") +
annotate("segment", x = 1, y = 0, xend = 1, yend = 1, linetype = "dotted") +
theme_classic() +
scale_x_continuous(breaks = c(0,1), labels = c("a", "b")) +
scale_y_continuous(breaks = c(0, 1), labels = c("0", "1/(b-a)")) +
labs(
  x = "x",
  y = "densité de probabilité (f(x))"
)

```



```

ggsave("img/03Rl09N.png", width = 14, height = 8, units = "cm")

```

3.4 Loi Binomiale

```

# choix de représenter entre 0 et 40
intervalle <- (0:40)

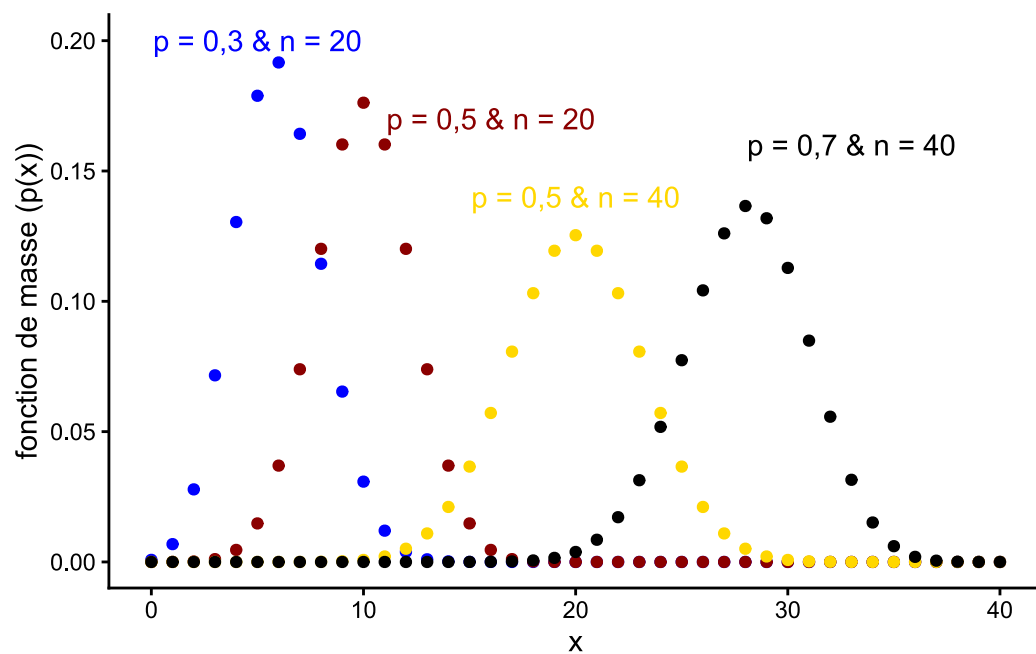
# loi binomiale avec p = 0.3, 0.5 ou 0.7 et pour 20 ou 40 épreuves de Bernoulli
tibble(
  loi_binomiale =
    rep(
      c(
        "p = 0,3 & n = 20",

```

```

      "p = 0,5 & n = 20",
      "p = 0,5 & n = 40",
      "p = 0,7 & n = 40"
    ),
    each = length(intervalle)
  ),
  x = rep(intervalle, 4),
  densite =
    c(
      dbinom(intervalle, prob = 0.3, size = 20),
      dbinom(intervalle, prob = 0.5, size = 20),
      dbinom(intervalle, prob = 0.5, size = 40),
      dbinom(intervalle, prob = 0.7, size = 40)
    )
) |>
ggplot() +
aes(x = x, y = densite, color = loi_binomiale) +
geom_point() +
theme_classic() +
scale_color_manual(values = c("blue", "darkred", "gold", "black")) +
annotate("text", x = 5, y = 0.2, label = "p = 0,3 & n = 20", color = "blue") +
  annotate("text", x = 16, y = 0.17, label = "p = 0,5 & n = 20", color =
"darkred") +
  annotate("text", x = 20, y = 0.14, label = "p = 0,5 & n = 40", color = "gold") +
  annotate("text", x = 33, y = 0.16, label = "p = 0,7 & n = 40") +
theme(legend.position = "none") +
labs(
  y = "fonction de masse (p(x))"
)

```



```
ggsave("img/03Rl10N.png", width = 14, height = 8, units = "cm")
```

3.5 Loi de Poisson

```
# choix de représenter entre 0 et 40
intervalle <- (0:40)

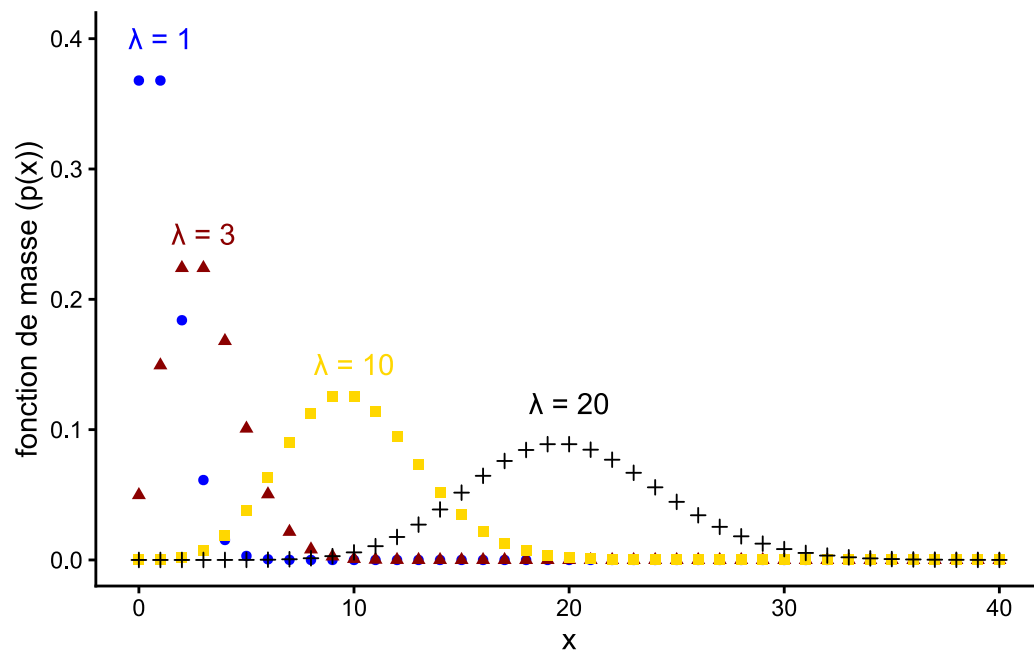
# loi de poisson de paramètre λ égale à 1, 3, 10 et 20
tibble(
  loi_de_poisson = factor(
    rep(
      c(
        "λ = 1",
        "λ = 3",
        "λ = 10",
        "λ = 20"
      ),
      each = length(intervalle)
    ),
    levels = c(
      "λ = 1",
      "λ = 3",
      "λ = 10",
      "λ = 20"
    )
  )
)
```



```

    )
  )
  ,
  x = rep(intervalle, 4),
  densite =
    c(
      dpois(intervalle, lambda = 1),
      dpois(intervalle, lambda = 3),
      dpois(intervalle, lambda = 10),
      dpois(intervalle, lambda = 20))
) |>
ggplot() +
aes(
  x = x,
  y = densite,
  color = loi_de_poisson,
  shape = loi_de_poisson
) +
geom_point() +
theme_classic() +
scale_color_manual(values = c("blue", "darkred", "gold", "black")) +
annotate("text", x = 1, y = 0.4, label = " $\lambda = 1$ ", color = "blue") +
annotate("text", x = 3, y = 0.25, label = " $\lambda = 3$ ", color = "darkred") +
annotate("text", x = 10, y = 0.15, label = " $\lambda = 10$ ", color = "gold") +
annotate("text", x = 20, y = 0.12, label = " $\lambda = 20$ ") +
ylab("fonction de masse (p(x))") +
theme(legend.position = "none")

```



```
ggsave("img/03Rl11N.png", width = 14, height = 8, units = "cm")
```

3.6 Loi normale

```
# choix de l'intervalle de représentation
intervalle <- seq(-15, 15, 0.1)

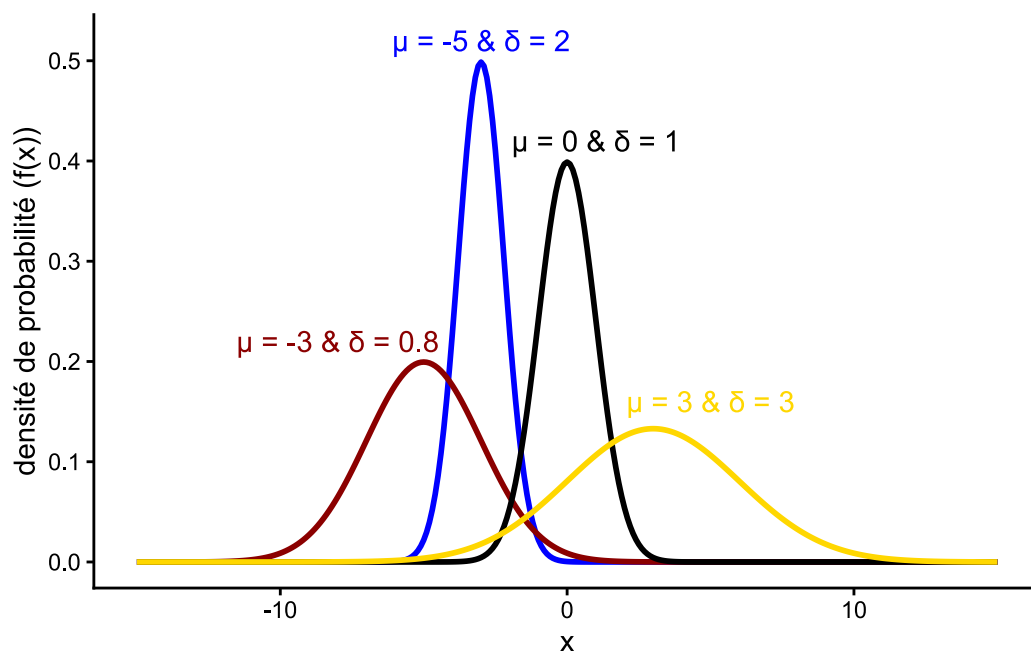
# loi normale de paramètres  $\mu$  égale à -5, -3, 0 ou 3 et  $\sigma$  égale à 2, 0.8, 1 ou 3
tibble(
  loi_normale =
    rep(
      c(
        " $\mu = 0$  &  $\sigma = 1$ ",
        " $\mu = 3$  &  $\sigma = 3$ ",
        " $\mu = -3$  &  $\sigma = 0.8$ ",
        " $\mu = -5$  &  $\sigma = 2$ "
      ),
      each = length(intervalle)
    ),
  x = rep(intervalle, 4),
  densite =
    c(
      dnorm(intervalle, mean = 0, sd = 1),
      dnorm(intervalle, mean = 3, sd = 3),

```

```

    dnorm(intervalle, mean = -3, sd = 0.8),
    dnorm(intervalle, mean = -5, sd = 2))
) |>
ggplot() +
  aes(x = x, y = densite, color = loi_normale) +
  geom_line(linewidth = 1) +
  theme_classic() +
  scale_color_manual(values = c("blue", "darkred", "black", "gold")) +
  annotate("text", x = -8, y = 0.22, label = "μ = -3 & \U03B4 = 0.8", color =
"darkred") +
  annotate("text", x = -3, y = 0.52, label = "μ = -5 & \U03B4 = 2", color =
"blue") +
  annotate("text", x = 1, y = 0.42, label = "μ = 0 & \U03B4 = 1") +
  annotate("text", x = 5, y = 0.16, label = "μ = 3 & \U03B4 = 3", color =
"gold") +
  ylab("densité de probabilité (f(x))") +
  theme(legend.position = "none")

```



```

ggsave("img/03Rl12N.png", width = 14, height = 8, units = "cm")

```

3.7 Loi exponentielle

```
# choix de l'intervalle de représentation
intervalle <- seq(0, 5, 0.01)

# loi exponentielle de paramètre  $\lambda$  égal à 0.5, 1, 3 et 5
tibble(
  loi_expo =
    rep(
      c(
        " $\lambda = 0.5$ ",
        " $\lambda = 1$ ",
        " $\lambda = 3$ ",
        " $\lambda = 5$ "
      ),
      each = length(intervalle)
    ),
  x = rep(intervalle, 4),
  densite =
    c(
      dexp(intervalle, rate = 0.5),
      dexp(intervalle, rate = 1),
      dexp(intervalle, rate = 3),
      dexp(intervalle, rate = 5)
    )
) |>
ggplot() +
  aes(x = x, y = densite, color = loi_expo) +
  geom_line(linewidth = 1) +
  annotate("text", x = 0.4, y = 4, label = " $\lambda = 5$ ", color = "gold") +
  annotate("text", x = 0.65, y = 1, label = " $\lambda = 3$ ") +
  annotate("text", x = 0.8, y = 0.7, label = " $\lambda = 1$ ", color = "darkred") +
  annotate("text", x = 3, y = 0.45, label = " $\lambda = 0.5$ ", color = "blue") +
  annotate("segment", x = log(2) * 1/5, y = dexp(log(2) * 1/5, rate = 5), xend =
log(2) * 1/5, yend = 0, linetype = "twodash", color = "gold", linewidth = 1) +
  annotate("segment", x = log(2) * 1/3, y = dexp(log(2) * 1/3, rate = 3), xend =
log(2) * 1/3, yend = 0, linetype = "twodash", color = "black", linewidth = 1) +
  theme_classic() +
  annotate("segment", x = log(2), y = dexp(log(2), rate = 1), xend = log(2),
yend = 0, linetype = "twodash", color = "dark red", linewidth = 1) +
  theme_classic() +
  annotate("segment", x = log(2) * 2, y = dexp(log(2) * 2, rate = 0.5), xend =
log(2) * 2, yend = 0, linetype = "twodash", color = "blue", linewidth = 1) +
  theme_classic() +
```



```
$ nom_eleve <glue> "Nom_1", "Nom_2", "Nom_3", "Nom_4", "Nom_5", "Nom_6", "No...
$ age_eleve <dbl> 8, 10, 9, 9, 9, 9, 9, 8, 7, 9, 8, 9, 9, 9, 10, 9, 9, 9, 9,...
```

```
summary(classe_fictive)
```

```
   groupe_eleve   nom_eleve   age_eleve
Min.   :1   Length   :30   Min.    : 7.000
1st Qu.:1   N.unique :30   1st Qu.: 9.000
Median :2   N.blank  : 0   Median : 9.000
Mean   :2   Min.nchar: 5   Mean    : 8.867
3rd Qu.:3   Max.nchar: 6   3rd Qu.: 9.000
Max.   :3                      Max.    :10.000
```

modification de la nature des données.

```
classe_fictive <-
  mutate(
    .data = classe_fictive,
    groupe_eleve = groupe_eleve |> as_factor(),
    nom_eleve = nom_eleve |> as.character()
  )

glimpse(classe_fictive)
```

```
Rows: 30
Columns: 3
$ groupe_eleve <fct> 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2, 3, 1, 2...
$ nom_eleve <chr> "Nom_1", "Nom_2", "Nom_3", "Nom_4", "Nom_5", "Nom_6", "Nom...
$ age_eleve <dbl> 8, 10, 9, 9, 9, 9, 9, 8, 7, 9, 8, 9, 9, 9, 10, 9, 9, 9, 9,...
```

4.2 variable qualitative

Il est conseillé d'aller voir la page d'aide sur le jeu de données avec `?mpg`.

```
# type de transmission (f = traction, r = propulsion, 4 = 4 roues motrices)
mpg |> count(drv)
```

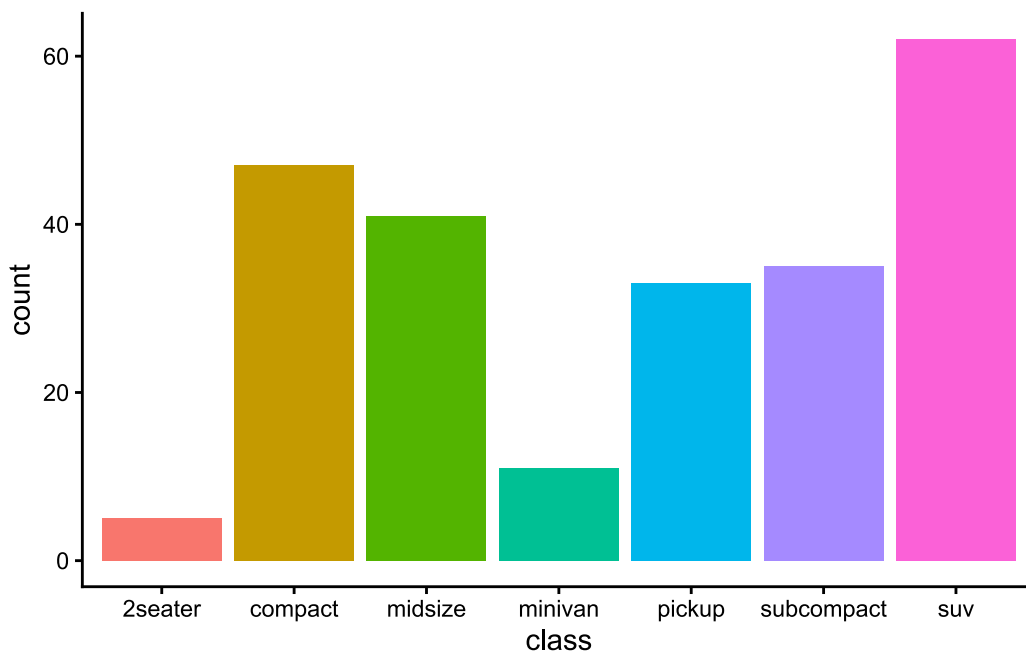
```
# A tibble: 3 × 2
  drv      n
<chr> <int>
1 f      4
2 r     103
```

```
2 f      106
3 r       25
```

```
# ajout d'une colonne pour calculer les proportions de chaque catégories
mpg |>
  count(manufacturer) |>
  mutate(prop = round(n*100/nrow(mpg))) |>
  arrange(desc(n)) |>
  write_csv(
    file = "data/table_resume_constructeur.csv"
  )
```

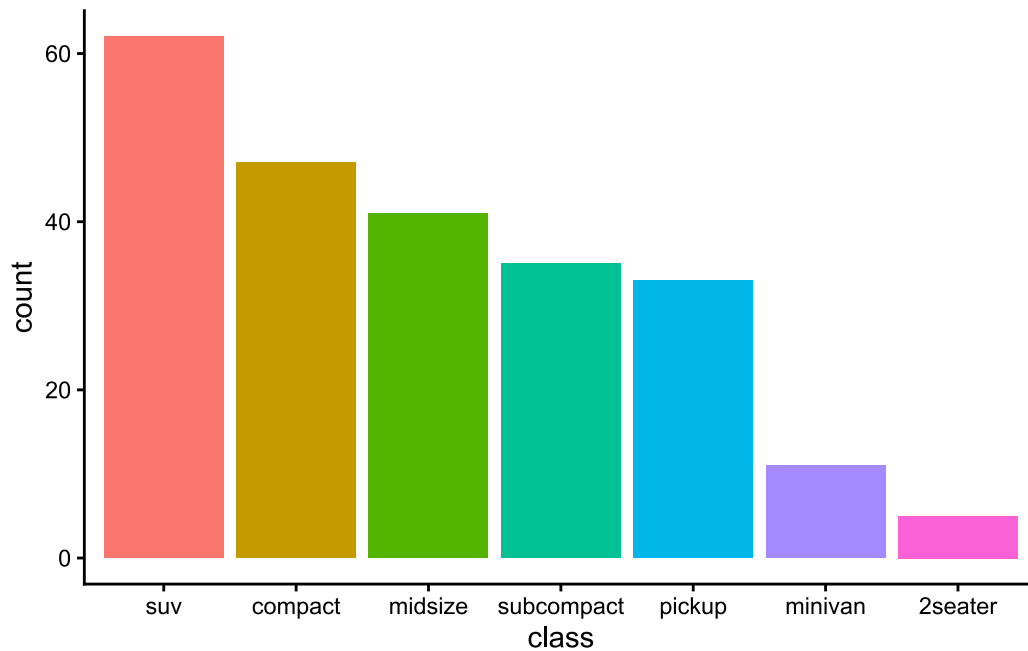
4.2.a création d'un graphique en barres

```
mpg |>
  ggplot() +
  aes(x = class, fill = class) +
  geom_bar() +
  theme_classic() +
  theme(legend.position = "none")
```



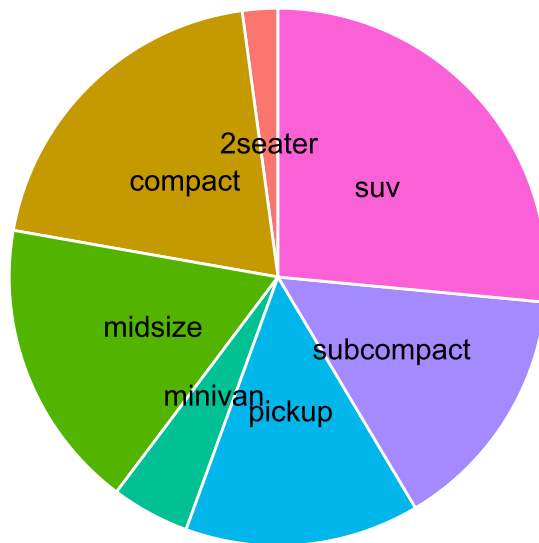
```
ggsave("img/03Rl18N.png", width = 14, height = 10, units = "cm")
```

```
mpg |>
  ggplot() +
  aes(x = class |> fct_infreq(), fill = class |> fct_infreq()) +
  labs(x = "class") +
  geom_bar() +
  theme_classic() +
  theme(legend.position = "none")
```



4.2.b création d'un graphique circulaire

```
mpg |>
  count(class) |>
  ggplot() +
  aes(x = "", y = n, fill = class) +
  geom_bar(stat = "identity", color = "white") +
  geom_text(aes(label = class), position = position_stack(vjust = 0.5)) +
  coord_polar("y") +
  theme_void() +
  theme(legend.position = "none")
```

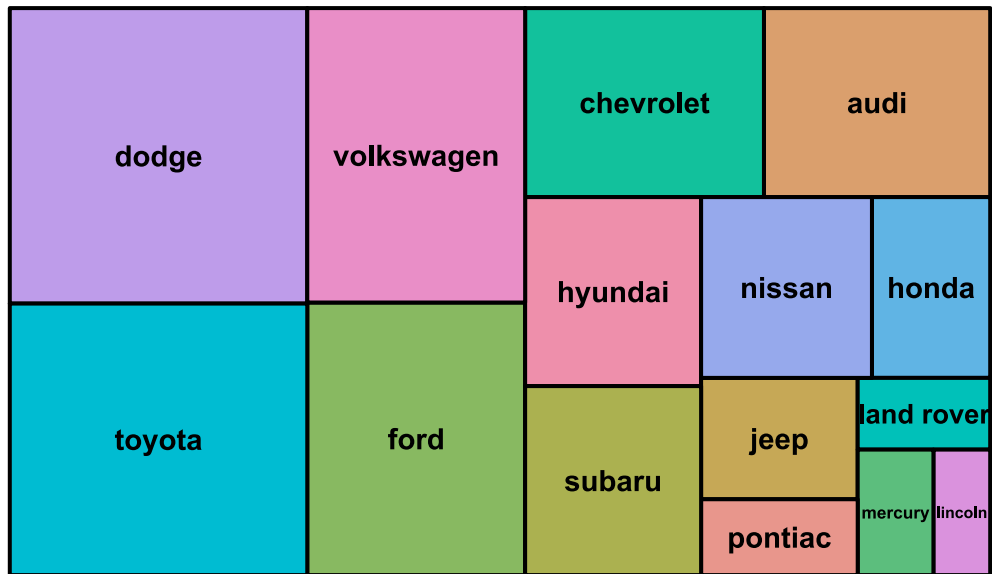



```
ggsave("img/03Rl19N.png", width = 10, height = 10, units = "cm")
```

4.2.c treemap

```
library(treemap)
treemap(
  mpg |>
    count(manufacturer),
  index = "manufacturer",
  vSize = "n",
  title = "Les constructeurs"
)
```

Les constructeurs



4.2.d nuage de mots

```
# création du tableau
tableau_constructeur <- mpg |>
  count(manufacturer)

# création du nuage de mots
wordcloud::wordcloud(
  words = tableau_constructeur$manufacturer,
  freq = tableau_constructeur$n,
  colors = c("gold", "blue", "dark red")
)
```



4.3 variable quantitative

4.3.a valeurs de dispersion

```
summary(mpg$cty)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
9.00	14.00	17.00	16.86	19.00	35.00

```
min(mpg$cty)
```

```
[1] 9
```

```
median(mpg$cty)
```

```
[1] 17
```

```
mean(mpg$cty)
```

```
[1] 16.85897
```

```
max(mpg$cty)
```

```
[1] 35
```

```
sd(mpg$cty)
```

```
[1] 4.255946
```

```
var(mpg$cty)
```

```
[1] 18.11307
```

```
#calcul de l'étendue  
range(mpg$cty) |> diff()
```

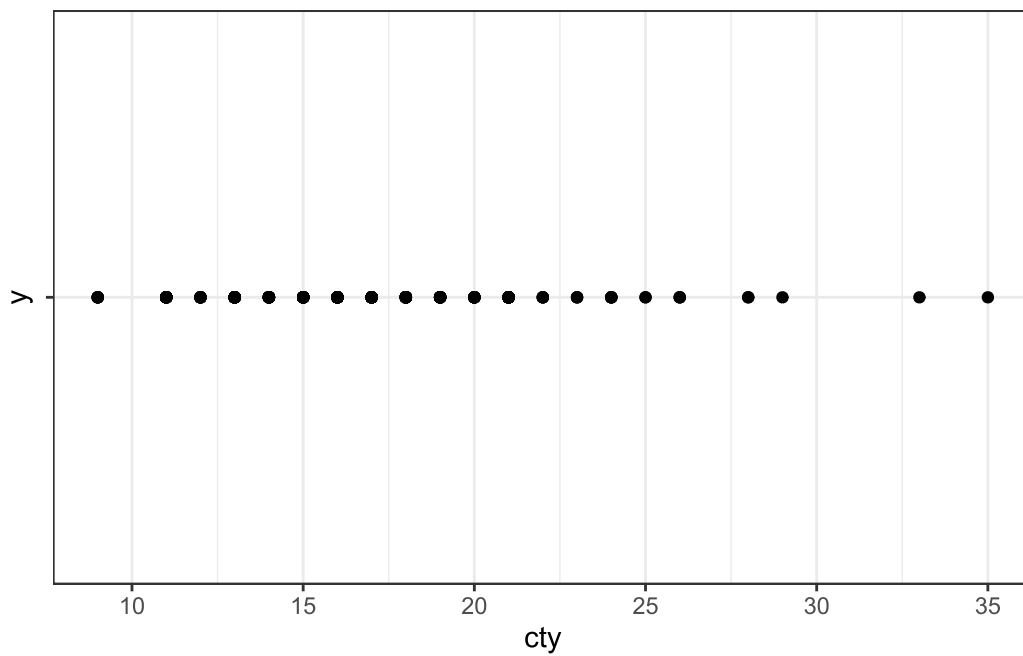
```
[1] 26
```

```
max(mpg$cty) - min(mpg$cty)
```

```
[1] 26
```

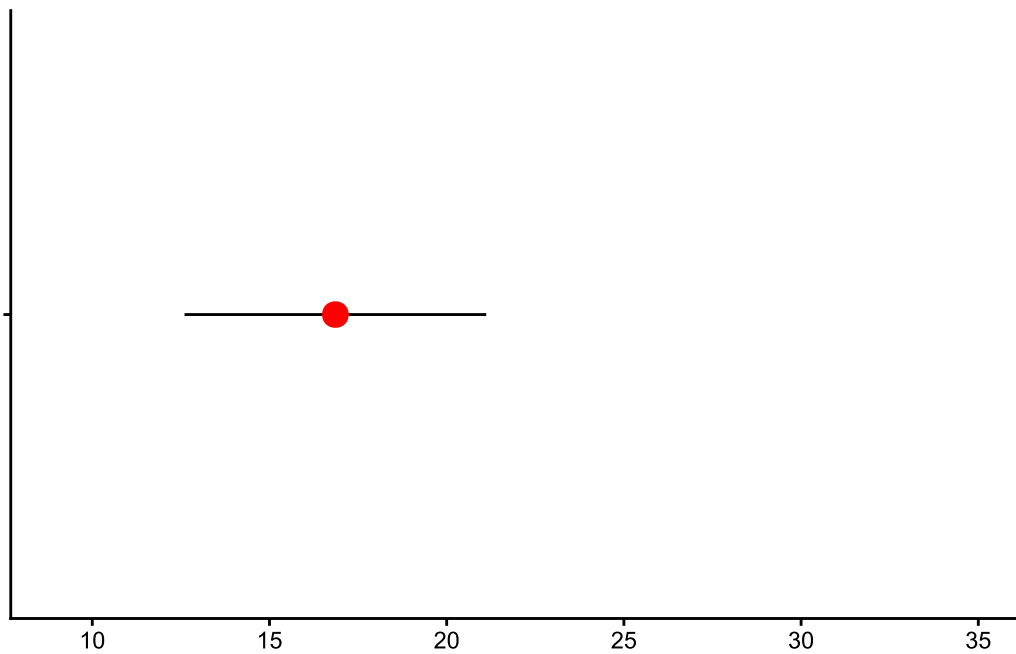
4.3.b nuage de points

```
ggplot(  
  data = mpg  
) +  
  aes(x = cty, y = "") +  
  geom_point() +  
  theme_bw()
```



4.3.c données de positionnement

```
mpg |>
  ggplot() +
  aes(x = cty, y = "") +
  annotate("segment", x = mean(mpg$cty) + sd(mpg$cty), xend = mean(mpg$cty) -
sd(mpg$cty), y = "", yend = "") +
  annotate("point", x = mean(mpg$cty), y = "", color = "red", size = 4) +
  xlim(9,35) +
  theme_classic()
```

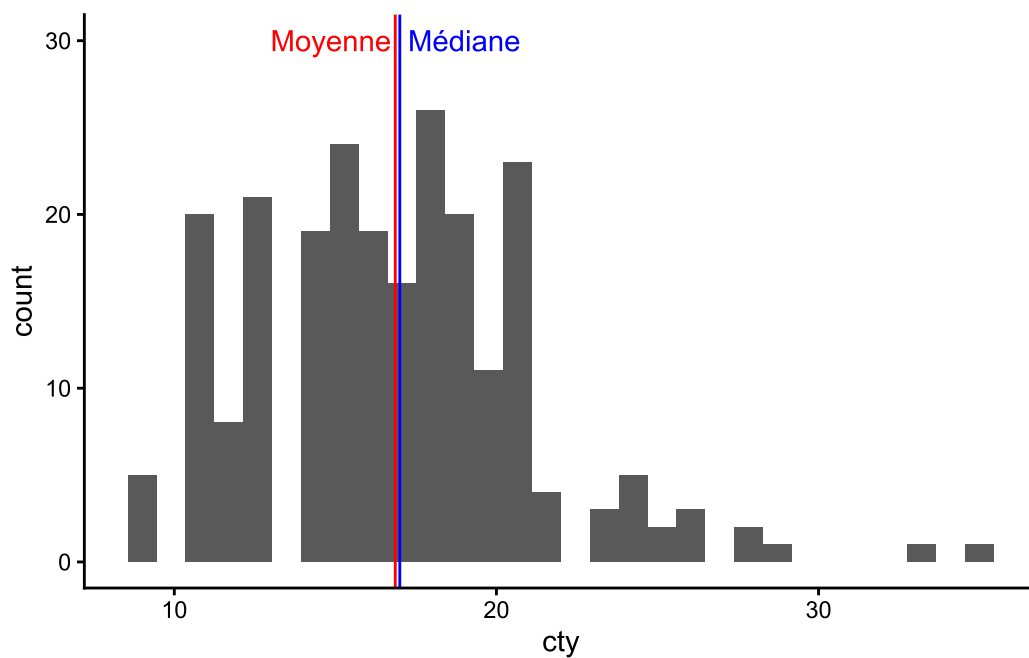


```
ggsave("img/03Rl24N.png", width = 14, height = 8, units = "cm")
```

4.3.d histogramme avec la moyenne et la médiane

```
ggplot(
  data = mpg
) +
  aes(x = cty) +
  geom_histogram() +
  geom_vline(aes(xintercept = mean(cty)), color = "red") +
  geom_vline(aes(xintercept = median(cty)), color = "blue") +
  annotate("text", x = mean(mpg$cty) - 2, y = 30, label = "Moyenne", color =
"red") +
  annotate("text", x = median(mpg$cty) + 2, y = 30, label = "Médiane", color =
"blue") +
  theme_classic()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



```
ggsave("img/03RL25N.png", width = 14, height = 8, units = "cm")
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

4.3.e Nuage de points avec la moyenne et la médiane

```
ggplot(iris) +
  aes(x = Petal.Length, y = "") +
  geom_point() +
  geom_point(aes(mean(Petal.Length)), color = "red", size = 4) +
  geom_point(aes(median(Petal.Length)), color = "blue", size = 4) +
  geom_text(
    aes(
      mean(Petal.Length),
      label = "Moyenne"
    ),
    nudge_y = 0.5,
    color = "red"
  ) +
  geom_text(
    aes(
      median(Petal.Length),
      label = "Médiane"
    )
  )
```

```
    ),  
    nudge_y = - 0.5,  
    color = "blue"  
  ) +  
  theme_classic()
```

Warning in `geom_point(aes(mean(Petal.Length)), color = "red", size = 4)`: All aesthetics have length 1, but the data has 150 rows.

i Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_point(aes(median(Petal.Length)), color = "blue", size = 4)`: All aesthetics have length 1, but the data has 150 rows.

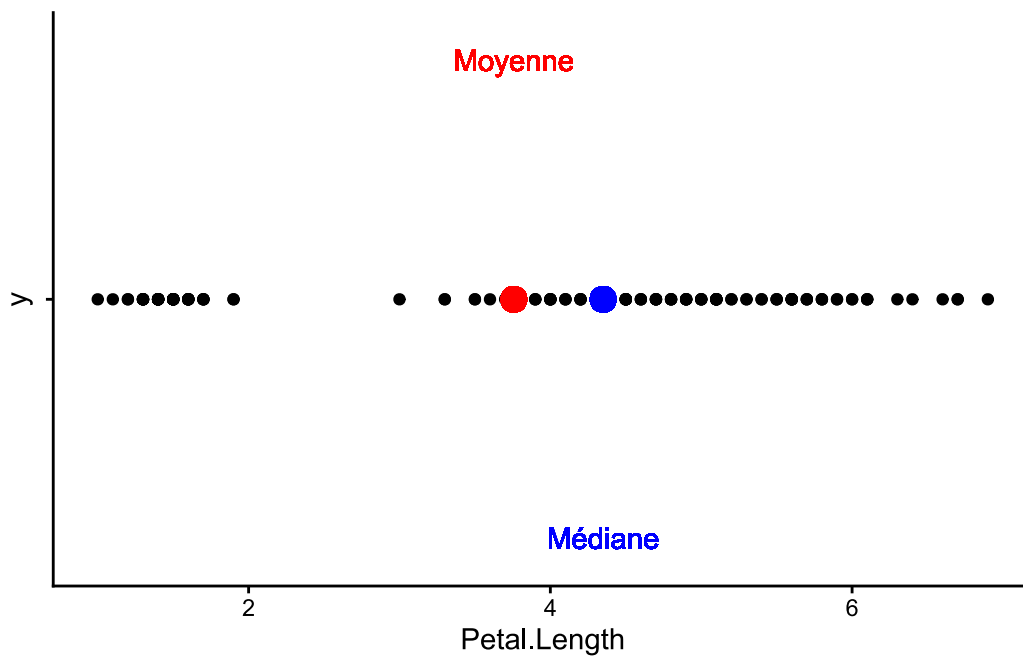
i Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_text(aes(mean(Petal.Length), label = "Moyenne"), nudge_y = 0.5, :` : All aesthetics have length 1, but the data has 150 rows.

i Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_text(aes(median(Petal.Length), label = "Médiane"), nudge_y = -0.5, :` : All aesthetics have length 1, but the data has 150 rows.

i Please consider using ``annotate()`` or provide this layer with data containing a single row.



4.3.f nuage de point dispersé

```
ggplot(
  data = iris
) +
  aes(x = Petal.Length, y = "") +
  geom_jitter() +
  geom_point(aes(mean(Petal.Length)), color = "red", size = 4) +
  geom_point(aes(median(Petal.Length)), color = "blue", size = 4) +
  geom_text(aes(mean(Petal.Length), label = "Moyenne"), nudge_y = 0.5, color =
"red") +
  geom_text(aes(median(Petal.Length), label = "Médiane"), nudge_y = - 0.5, color
= "blue") +
  theme_classic()
```

Warning in `geom_point(aes(mean(Petal.Length)), color = "red", size = 4)`: All aesthetics have length 1, but the data has 150 rows.
i Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_point(aes(median(Petal.Length)), color = "blue", size = 4)`: All aesthetics have length 1, but the data has 150 rows.

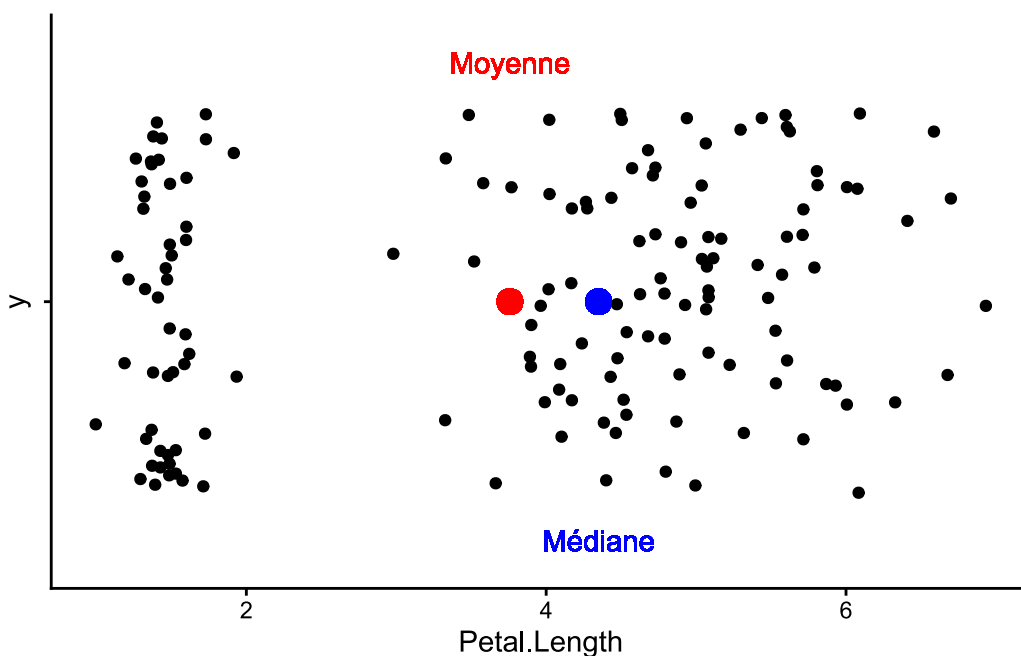
i Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_text(aes(mean(Petal.Length), label = "Moyenne"), nudge_y = 0.5, :`
All aesthetics have length 1, but the data has 150 rows.

i Please consider using ``annotate()`` or provide this layer with data containing a single row.

Warning in `geom_text(aes(median(Petal.Length), label = "Médiane"), nudge_y =`
-0.5, : All aesthetics have length 1, but the data has 150 rows.

i Please consider using ``annotate()`` or provide this layer with data containing a single row.



```
iris |>
  count(Petal.Length) |>
  ggplot() +
  aes(x = Petal.Length, y = "", size = n) +
  geom_point() +
  geom_point(aes(weighted.mean(Petal.Length, n), y = ""), color = "red",
inherit.aes = FALSE) +
  geom_point(aes(matrixStats::weightedMedian(Petal.Length, n), y = ""), color =
"blue", inherit.aes = FALSE) +
```

```

    geom_text(aes(weighted.mean(Petal.Length, n), y = "", label = "Moyenne
pondérée"), nudge_y = 0.5, color = "red", inherit.aes = FALSE) +
    geom_text(aes(matrixStats::weightedMedian(Petal.Length, n), y = "", label =
"Médiane pondérée"), nudge_y = - 0.5, color = "blue", inherit.aes = FALSE) +
    theme_classic()

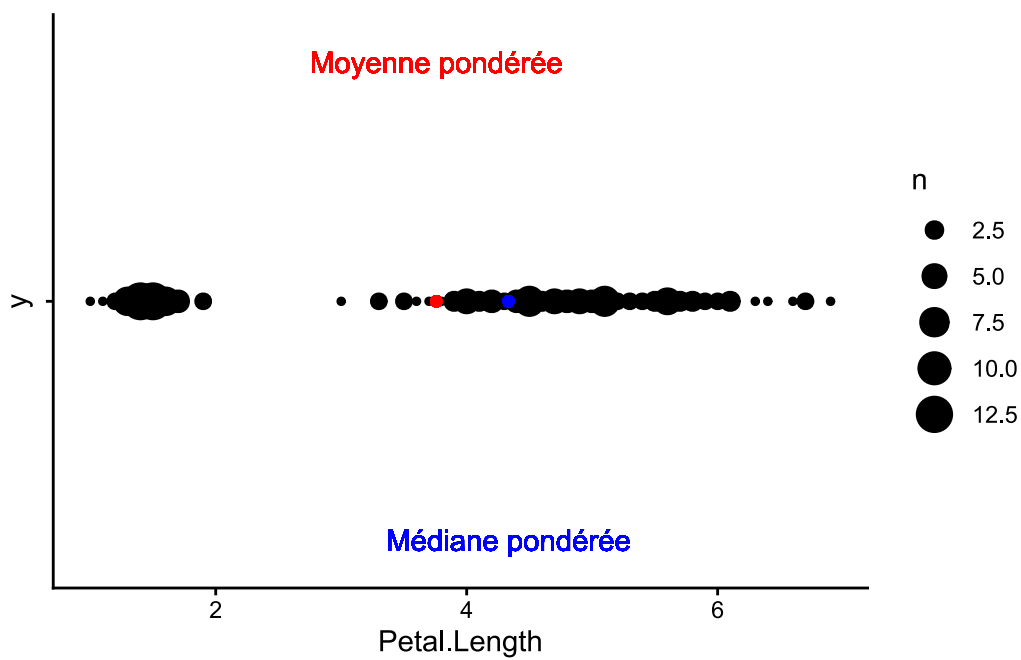
```

Warning in geom_point(aes(weighted.mean(Petal.Length, n), y = ""), color = "red", : All aesthetics have length 1, but the data has 43 rows.
i Please consider using `annotate()` or provide this layer with data containing a single row.

Warning in geom_point(aes(matrixStats::weightedMedian(Petal.Length, n), : All aesthetics have length 1, but the data has 43 rows.
i Please consider using `annotate()` or provide this layer with data containing a single row.

Warning in geom_text(aes(weighted.mean(Petal.Length, n), y = "", label = "Moyenne pondérée"), : All aesthetics have length 1, but the data has 43 rows.
i Please consider using `annotate()` or provide this layer with data containing a single row.

Warning in geom_text(aes(matrixStats::weightedMedian(Petal.Length, n), y = "", : All aesthetics have length 1, but the data has 43 rows.
i Please consider using `annotate()` or provide this layer with data containing a single row.



4.3.g quantiles

```
quantile(x = mpg$cty)
```

0%	25%	50%	75%	100%
9	14	17	19	35

```
# déciles
quantile(
  x = mpg$cty,
  probs = seq(0, 1, 0.1)
)
```

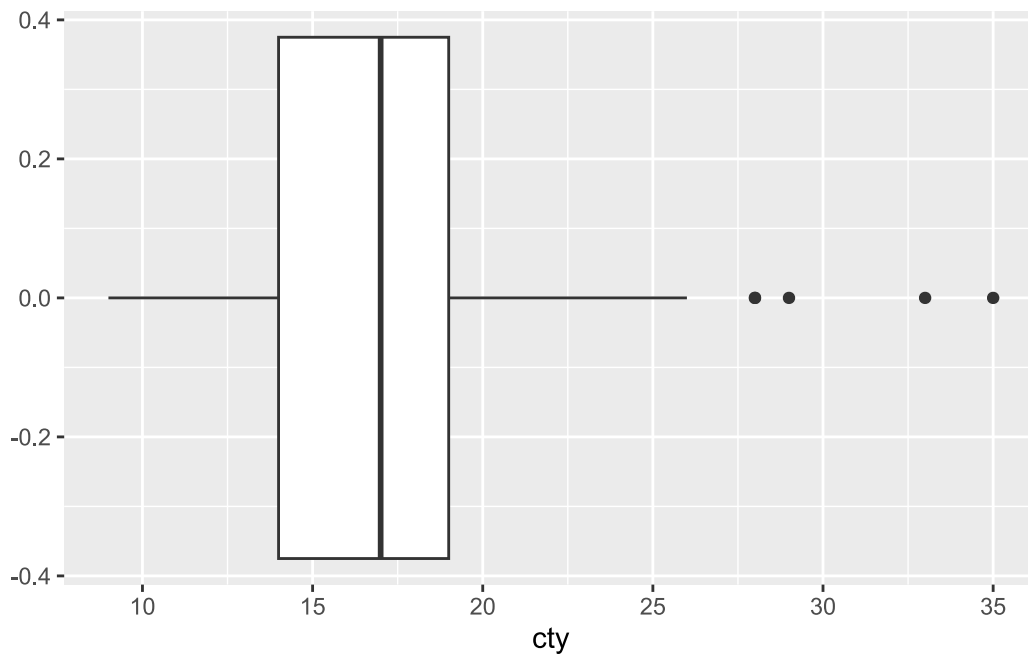
0%	10%	20%	30%	40%	50%	60%	70%	80%	90%	100%
9	11	13	14	15	17	18	19	20	21	35

```
# centiles
centile <- quantile(
  x = mpg$cty,
  probs = seq(0, 1, 0.01)
)
centile
```

0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	11%	12%
9.00	9.00	10.32	11.00	11.00	11.00	11.00	11.00	11.00	11.00	11.00	12.00	12.00
13%	14%	15%	16%	17%	18%	19%	20%	21%	22%	23%	24%	25%
12.00	12.62	13.00	13.00	13.00	13.00	13.00	13.00	13.00	13.00	13.59	14.00	14.00
26%	27%	28%	29%	30%	31%	32%	33%	34%	35%	36%	37%	38%
14.00	14.00	14.00	14.00	14.00	14.23	15.00	15.00	15.00	15.00	15.00	15.00	15.00
39%	40%	41%	42%	43%	44%	45%	46%	47%	48%	49%	50%	51%
15.00	15.00	15.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	17.00	17.00
52%	53%	54%	55%	56%	57%	58%	59%	60%	61%	62%	63%	64%
17.00	17.00	17.00	17.00	17.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00	18.00
65%	66%	67%	68%	69%	70%	71%	72%	73%	74%	75%	76%	77%
18.00	18.00	18.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.00	19.08	20.00
78%	79%	80%	81%	82%	83%	84%	85%	86%	87%	88%	89%	90%
20.00	20.00	20.00	20.73	21.00	21.00	21.00	21.00	21.00	21.00	21.00	21.00	21.00
91%	92%	93%	94%	95%	96%	97%	98%	99%	100%			
22.00	22.00	23.00	24.00	24.00	24.68	26.00	26.68	28.67	35.00			

4.3.h boîte à moustaches

```
# simple
ggplot(data = mpg) +
  geom_boxplot(aes(cty))
```



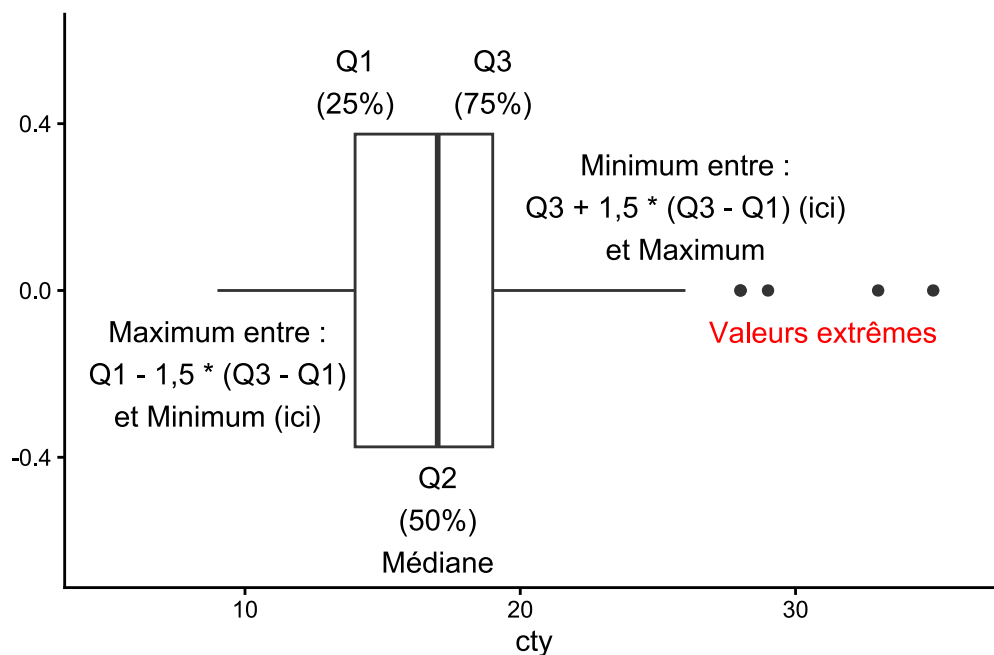
```

# avec commentaires
quartile <- quantile(mpg$cty)

ggplot(data = mpg) +
  geom_boxplot(aes(cty)) +
  annotate(
    "text",
    x = quartile[1],
    y = - 0.2 ,
    label = "Maximum entre : \nQ1 - 1,5 * (Q3 - Q1)\net Minimum (ici)"
  ) +
  annotate(
    "text",
    x = quartile[2],
    y = 0.5,
    label = "Q1\n(25%)"
  ) +
  annotate(
    "text",
    x = quartile[3],
    y = - 0.55,
    label = "Q2\n(50%)\nMédiane"
  ) +
  annotate(
    "text",
    x = quartile[4],
    y = 0.5,
    label = "Q3\n(75%)"
  ) +
  annotate(
    "text",
    x = 26,
    y = 0.2,
    label = "Minimum entre : \nQ3 + 1,5 * (Q3 - Q1) (ici)\net Maximum" ) +
  annotate(
    "text",
    x = 31,
    y = -0.1,
    label = "Valeurs extrêmes",
    color = "red"
  ) +
  xlim(5, 36) +
  ylim(-0.65, 0.6) +

```

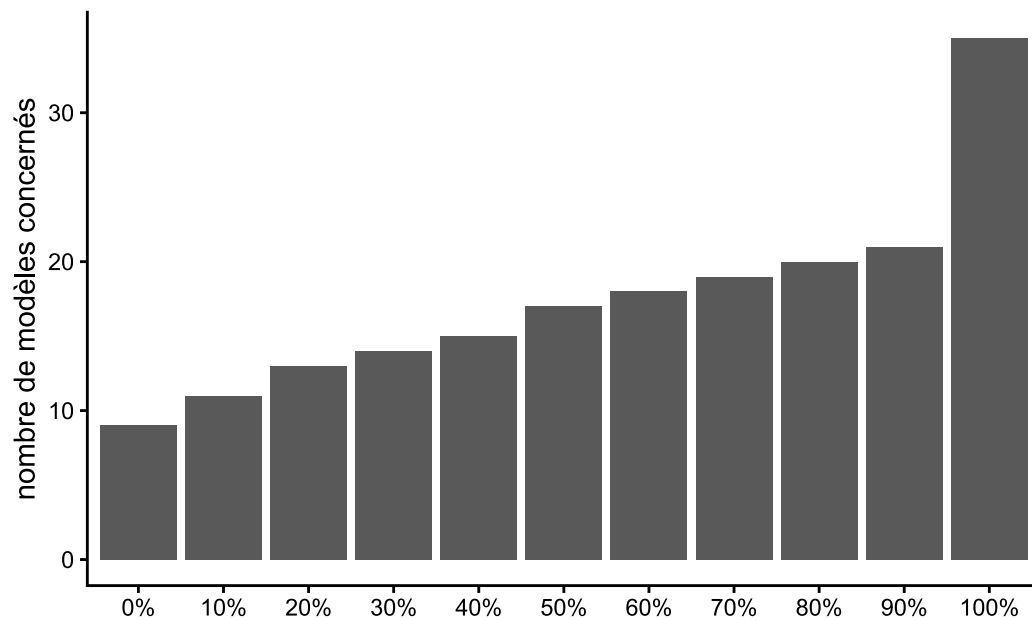
```
ylab("") +  
theme_classic()
```



4.3.i visualisation des deciles

```
cty_decile <- quantile(  
  x = mpg$cty,  
  probs = seq(0, 1, 0.1)  
)  
  
tibble(  
  pourcentage = names(cty_decile),  
  cty_deciles = cty_decile  
) |>  
ggplot() +  
aes(  
  x = pourcentage |> fct_inorder(),  
  y = cty_decile  
) +  
geom_bar(stat = "identity") +  
labs(  
  x = "",  
  y = "nombre de modèles concernés"
```

```
) +  
theme_classic()
```



```
ggsave("img/03Rl31N.png", width = 14, height = 8, units = "cm")
```

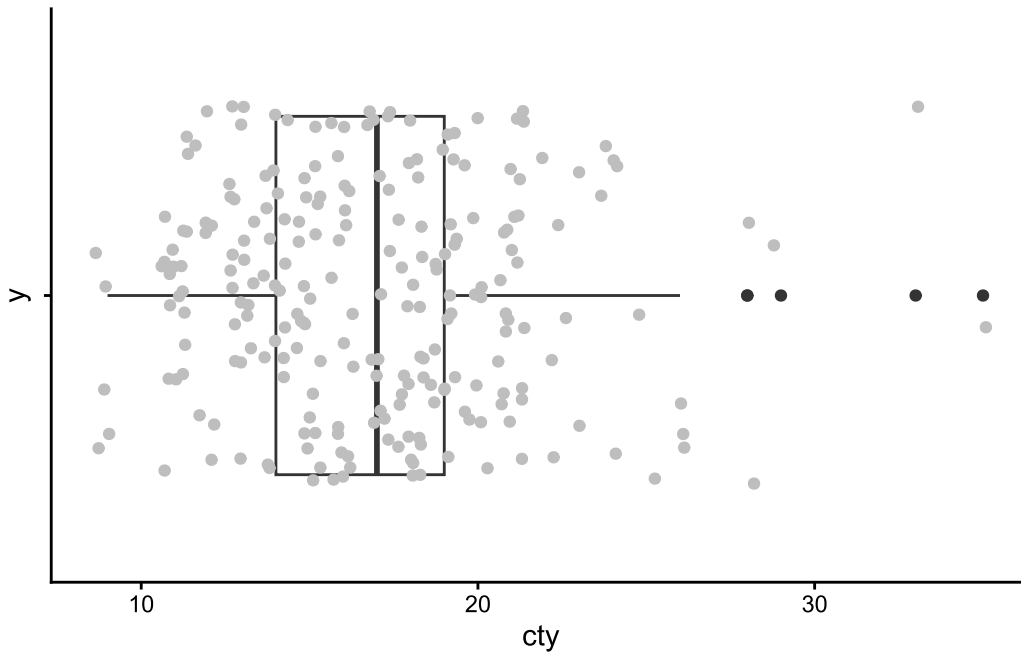
4.3.j filtrer sur les valeurs extremes

```
mpg |>  
  filter(  
    cty > 26  
  )
```

```
# A tibble: 5 × 11  
  manufacturer model    displ  year   cyl trans  drv    cty   hwy fl    class  
    <chr>         <chr>    <dbl> <int> <int> <chr> <chr> <int> <int> <chr> <chr>  
1 honda         civic      1.6  1999     4 manua... f      28    33 r    subc...  
2 toyota        corolla    1.8  2008     4 manua... f      28    37 r    comp...  
3 volkswagen    jetta      1.9  1999     4 manua... f      33    44 d    comp...  
4 volkswagen    new beetle  1.9  1999     4 manua... f      35    44 d    subc...  
5 volkswagen    new beetle  1.9  1999     4 auto(... f      29    41 d    subc...
```


4.3.k boîte à moustaches avec points dispersés

```
ggplot(data = mpg) +  
  aes(x= cty, y = "") +  
  geom_boxplot() +  
  geom_jitter(color = "grey") +  
  theme_classic()
```



5 Statistiques descriptives bivariées

5.1 2 variables quantitatives

5.1.a test de Pearson

```
cor.test(mpg$cty, mpg$hwy)
```

Pearson's product-moment correlation

data: mpg\$cty and mpg\$hwy

t = 49.585, df = 232, p-value < 2.2e-16

alternative hypothesis: true correlation is not equal to 0

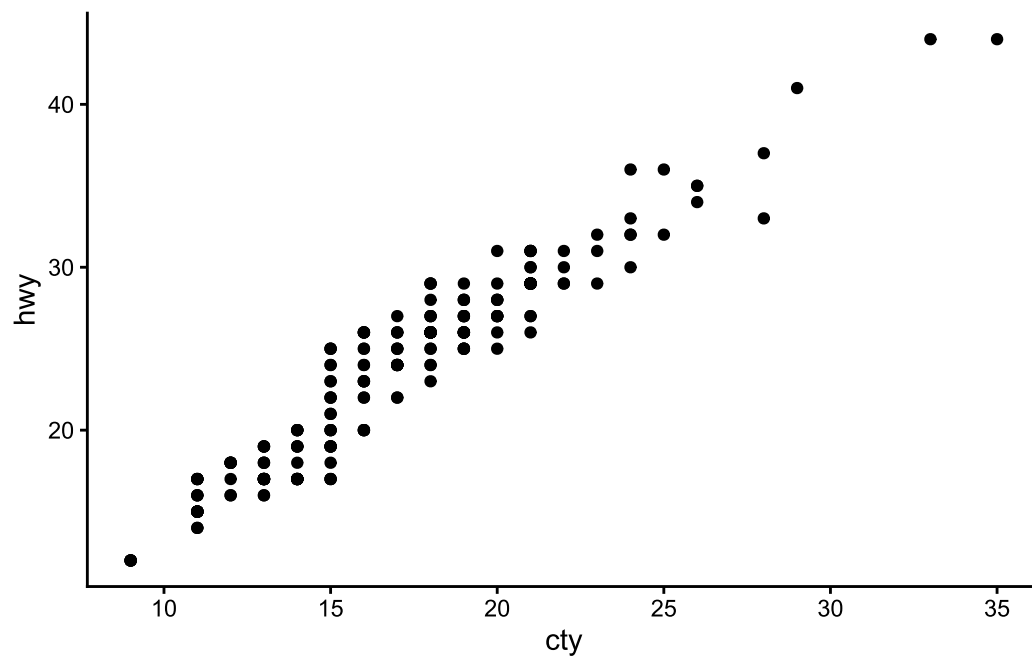
95 percent confidence interval:

0.9433129 0.9657663

```
sample estimates:
      cor 
0.9559159
```

5.1.b graphique

```
ggplot(mpg) +
  aes(
    x = cty,
    y = hwy
  ) +
  geom_point() +
  theme_classic()
```



5.2 Une quanti et une quali

5.2.a test du Khi-2

```
chisq.test(mpg$cty, mpg$class)
```

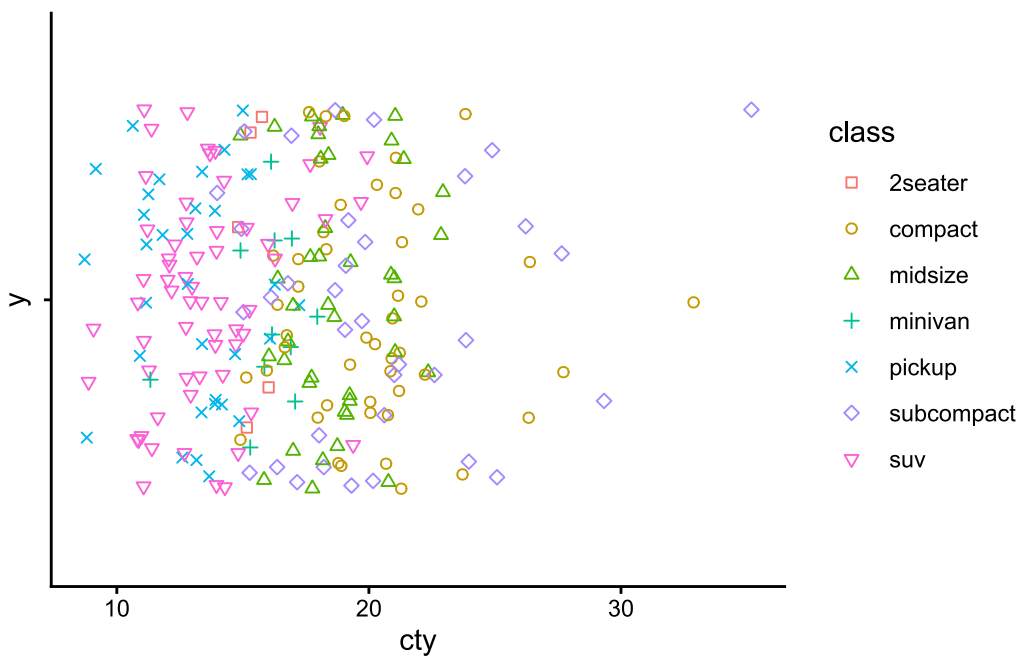
```
Warning in chisq.test(mpg$cty, mpg$class): L'approximation du Chi-2 est
peut-être incorrecte
```

Pearson's Chi-squared test

```
data: mpg$cty and mpg$class  
X-squared = 293.17, df = 120, p-value < 2.2e-16
```

5.2.b représentation graphique avec forme et couleur

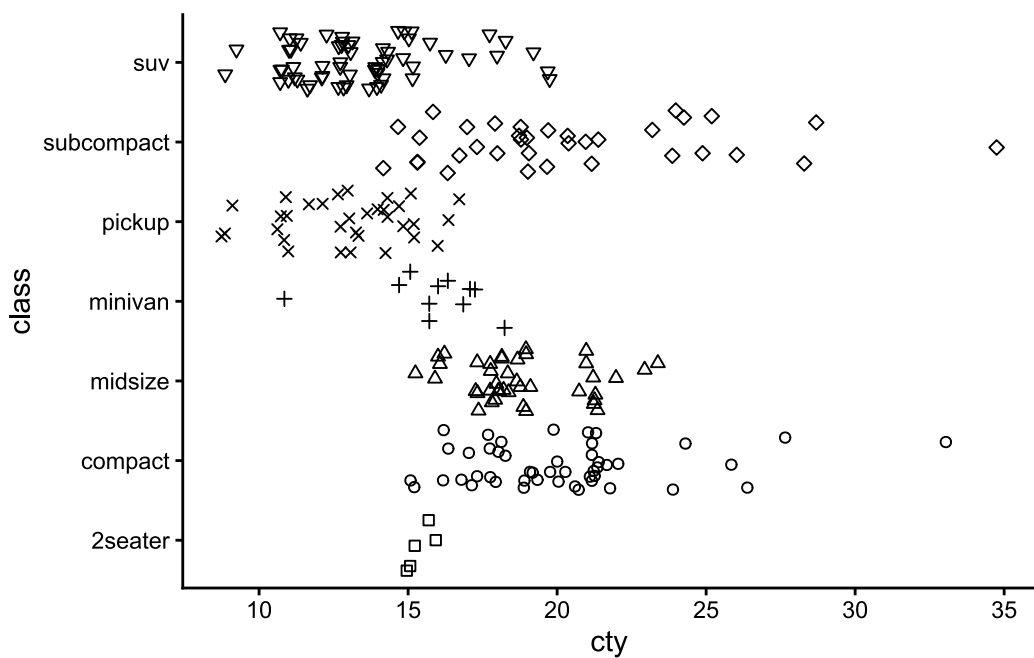
```
ggplot(mpg) +  
  aes(x = cty, y = "", color = class, shape = class) +  
  geom_jitter() +  
  scale_shape_manual(values = c(0:6)) +  
  theme_classic()
```



```
ggsave("img/03Rl35N.png", width = 14, height = 8, units = "cm")
```

5.2.c une ligne par type de véhicule

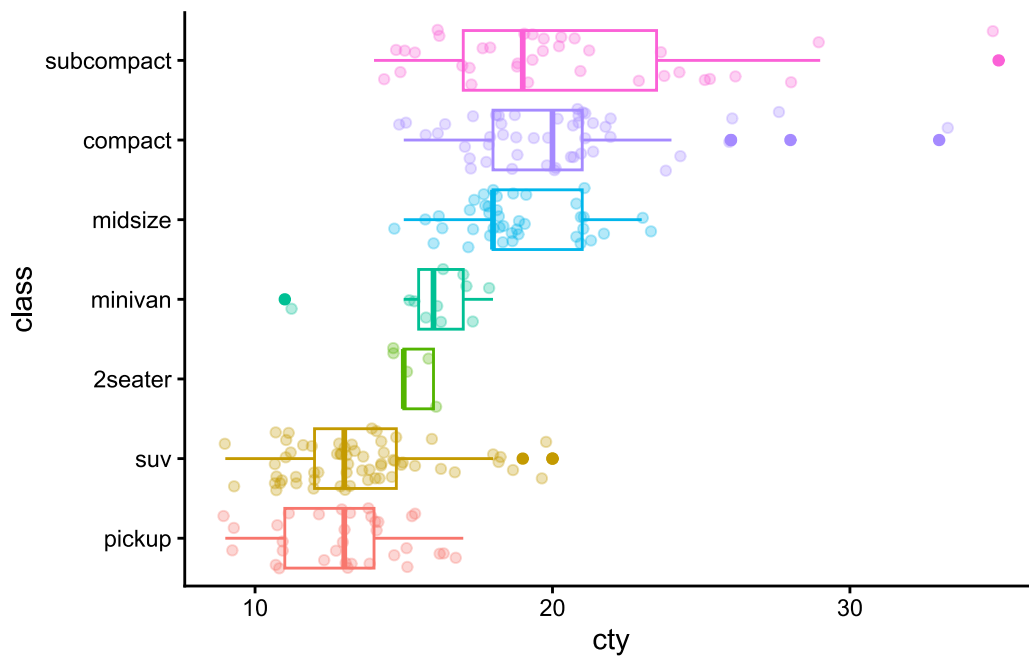
```
ggplot(mpg) +  
  aes(x = cty, y = class, shape = class) +  
  geom_jitter() +  
  scale_shape_manual(values = c(0:6)) +  
  theme_classic() +  
  theme(legend.position = "none")
```



```
ggsave("img/03RL36N.png", width = 14, height = 8, units = "cm")
```

5.2.d avec des boîtes à moustaches

```
ggplot(mpg) +
  aes(x = cty, y = reorder(class, cty), color = reorder(class, cty)) +
  geom_boxplot() +
  geom_jitter(alpha = 0.3) +
  labs(
    y = "class",
    color = "class"
  ) +
  theme_classic() +
  theme(legend.position = "none")
```



```
ggsave("img/03RL37N.png", width = 14, height = 8, units = "cm")
```

5.3 2 variables qualitatives

5.3.a tableau de contingence

```
mpg |>
  count(manufacturer, class) |>
  pivot_wider(names_from = class, values_from = n)
```

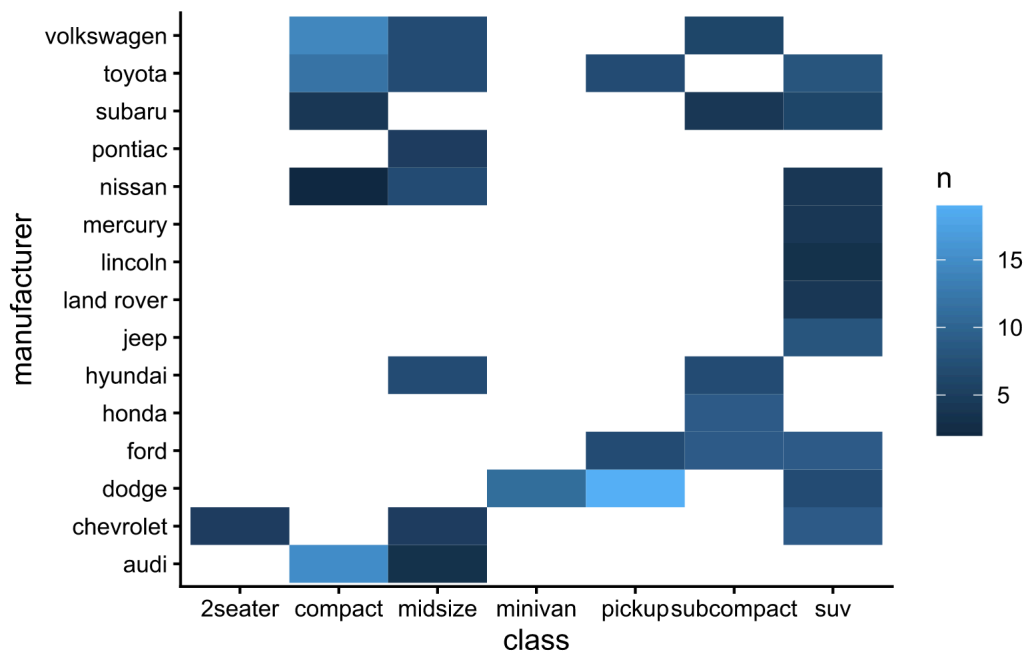
```
# A tibble: 15 × 8
```

	manufacturer	compact	midsize	`2seater`	suv	minivan	pickup	subcompact
	<chr>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
1	audi	15	3	NA	NA	NA	NA	NA
2	chevrolet	NA	5	5	9	NA	NA	NA
3	dodge	NA	NA	NA	7	11	19	NA
4	ford	NA	NA	NA	9	NA	7	9
5	honda	NA	NA	NA	NA	NA	NA	9
6	hyundai	NA	7	NA	NA	NA	NA	7
7	jeep	NA	NA	NA	8	NA	NA	NA
8	land rover	NA	NA	NA	4	NA	NA	NA
9	lincoln	NA	NA	NA	3	NA	NA	NA
10	mercury	NA	NA	NA	4	NA	NA	NA
11	nissan	2	7	NA	4	NA	NA	NA

12	pontiac	NA	5	NA	NA	NA	NA	NA
13	subaru	4	NA	NA	6	NA	NA	4
14	toyota	12	7	NA	8	NA	7	NA
15	volkswagen	14	7	NA	NA	NA	NA	6

5.3.b carte des points chauds

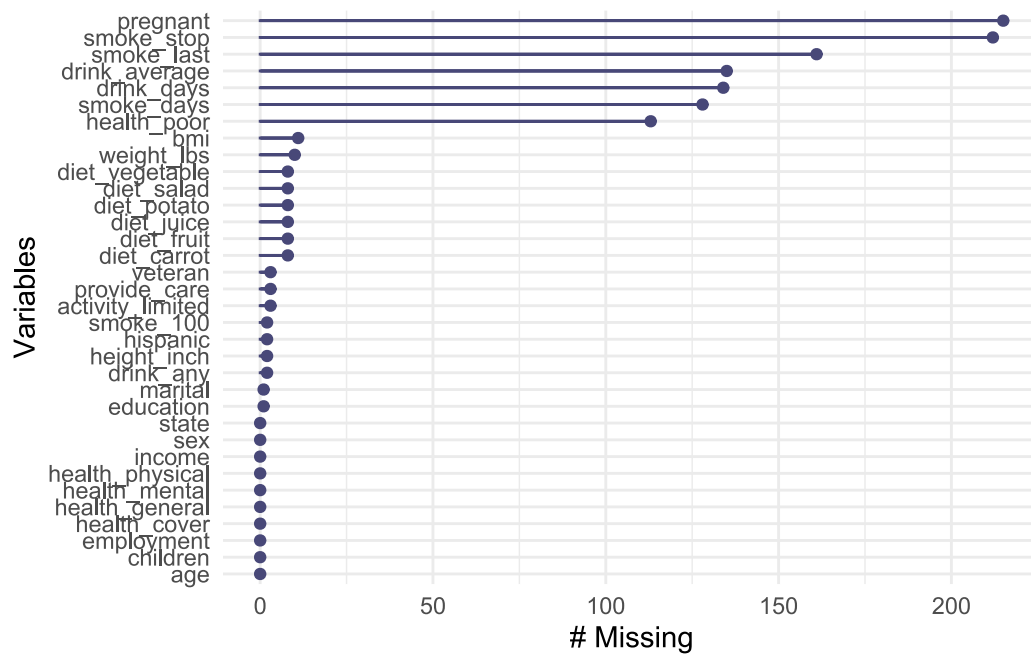
```
mpg |>
  count(manufacturer, class) |>
  ggplot() +
  aes(x = class, y = manufacturer, fill = n) +
  geom_tile() +
  theme_classic()
```



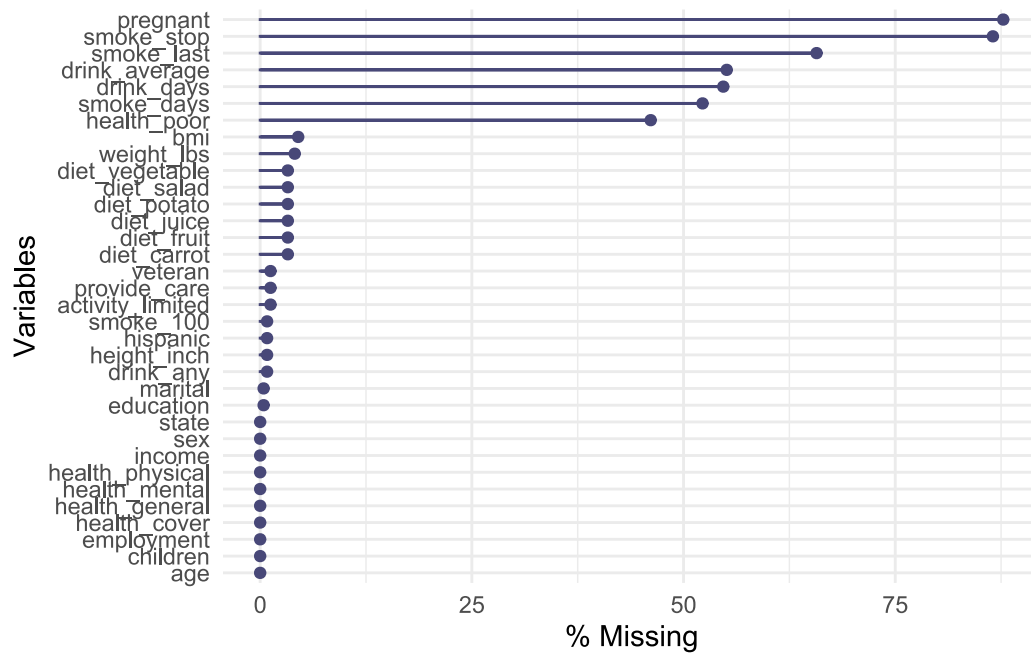
6 Valeurs manquantes

```
library(naniar)

gg_miss_var(riskfactors)
```



```
gg_miss_var(riskfactors, show_pct = TRUE)
```



```
gg_miss_upset(riskfactors)
```

Warning: `aes\_string()` was deprecated in ggplot2 3.0.0.
 i Please use tidy evaluation idioms with `aes()`.
 i See also `vignette("ggplot2-in-packages")` for more information.
 i The deprecated feature was likely used in the UpSetR package.
 Please report the issue at <<https://github.com/hms-dbmi/UpSetR/issues>>.

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.
 i The deprecated feature was likely used in the UpSetR package.
 Please report the issue at <<https://github.com/hms-dbmi/UpSetR/issues>>.

Warning: The `size` argument of `element\_line()` is deprecated as of ggplot2 3.4.0.
 i Please use the `linewidth` argument instead.
 i The deprecated feature was likely used in the UpSetR package.
 Please report the issue at <<https://github.com/hms-dbmi/UpSetR/issues>>.

